

A Verification Pearl: Differential Uniformity, Done Once

Abstract

We give one small definition.

We prove one small closure law.

We get the Frobenius twist for free.

Everything is checked in Lean 4 / Mathlib.

Conference paper call — short, slow-read submission.

1 The idea

A map f is *APN* if its derivatives are at most two-to-one.

The literature writes this only in characteristic two.

We write it once, for any additive group.

One number. No field. No characteristic.

2 The primitive

The discrete derivative:

$$\text{derivMap } f \ a \ x \ = \ f(x + a) - f(x).$$

Its fibre size:

$$\text{fiberCard } f \ a \ b \ = \ \#\{x : \text{derivMap } f \ a \ x = b\}.$$

The differential uniformity:

$$\delta(f) \ = \ \max_{a \neq 0, b} \text{fiberCard } f \ a \ b.$$

$$\text{IsAPN } f \ : \Longleftrightarrow \ \delta(f) = 2.$$

$$\text{IsPN } f \ : \Longleftrightarrow \ \delta(f) = 1.$$

3 The primitive, in Lean

```
def derivMap (f : V → W) (a x : V) : W := f (x + a) - f x

def fiberCard (f : V → W) (a : V) (b : W) : ℕ :=
  (univ.filter fun x => derivMap f a x = b).card

def differentialUniformity (f : V → W) : ℕ :=
  (univ.filter fun a => a ≠ 0).sup
    fun a => univ.sup fun b => fiberCard f a b

def IsAPN (f : V → W) : Prop := differentialUniformity f = 2
```

V, W are any finite additive groups.

4 The closure law

Theorem 1. *Let σ be an additive bijection.*

$$\delta(\sigma \circ f) = \delta(f).$$

Why: σ commutes with the derivative, so it just relabels fibres.

$$\text{derivMap}(\sigma \circ f) \, a \, x = \sigma(\text{derivMap} \, f \, a \, x).$$

So APN is preserved.

$$\text{IsAPN} \, f \implies \text{IsAPN}(\sigma \circ f).$$

5 The closure law, in Lean

```
lemma differentialUniformity_comp_addEquiv
  (f : V → W) σ( : W ≃+ W') :
  differentialUniformity σ( ∘ f) = differentialUniformity f

theorem IsAPN.comp_addEquiv
  {f : V → W} (hf : IsAPN f) σ( : W ≃+ W') :
  IsAPN σ( ∘ f)
```

This moves a whole EA-class at once.

6 Application: the Frobenius twist

Now take a finite field of characteristic two.

The map

$$\sigma : x \mapsto x^{2^j}$$

is an additive bijection.

So it is a special case of the closure law.

Corollary 1. *If f is APN, then $x \mapsto f(x)^{2^j}$ is APN.*

7 The twist, in Lean

```
theorem IsAPN.comp_frobenius
  {f : F → F} (hf : IsAPN f) (j : ℕ) :
  IsAPN (fun x => (f x) ^ (2 ^ j))
```

The same move drives the Kasami extension to even k :

```
lemma apn_frob_twist {d : ℕ}
  (hd : IsAPN (fun x : F => x ^ d)) (j : ℕ) :
  IsAPN (fun x : F => x ^ (d * 2 ^ j))
```

Kasami even k = Kasami odd k , twisted.

8 The bridge back

In characteristic two the old form returns as a corollary.

Theorem 2. *If $\delta(f) \leq 2$ then, for $a \neq 0$,*

$$f(x+a) - f(x) = f(y+a) - f(y) \implies y = x \vee y = x + a.$$

Nothing is lost. The classical statement is derived, not assumed.

9 What we claim

- One characteristic-free definition.
- One closure law.
- Frobenius twist as a one-line corollary.
- Fully machine-checked.

Read slowly. There is not much to read.